

# Estimating the causal effect of replying “👍” versus “merci” to professional emails

Constance Reply-All  
Department of Organizational Communication  
Institute for Workplace Efficiency  
Inbox City, DE  
c.reply-all@example.org

Marcus Cc  
Behavioral Email Analytics Lab  
University of Inbox Zero  
Threadford, NJ  
marcus.cc@example.org

**Abstract—Background:** professional emails increasingly close with either a written sign-off or a bare emoji reaction, and whether this choice causally affects downstream communication outcomes is unknown. **Methods:** we analyzed metadata from 1,204 professional email threads across six organizations, comparing threads whose final message closed with a “👍” emoji, the word “merci,” or no sign-off, using inverse-probability-weighted estimates of the probability of a further reply within 48 hours. **Results:** reply probability differed by sign-off group both before and after adjustment for measured confounders. **Conclusion:** these findings are consistent with sign-off choice being causally relevant to reply behavior in professional email correspondence.

**Index Terms**—Causal inference, Email etiquette, Digital communication, Propensity score

## Summary of changes

| Comment  | Type        | Page | Section      |
|----------|-------------|------|--------------|
| R1-1     | addition    | 1    | Introduction |
| R1-2     | replacement | 2    | Methods      |
| R1-3     | replacement | 2    | Methods      |
| R1-5     | replacement | 2    | Methods      |
| R1-6     | deletion    | 2    | Methods      |
| R1-7     | addition    | 3    | Results      |
| R1-9     | suppression | 3    | Results      |
| R1-10    | addition    | 1    | Introduction |
| R1-10    | addition    | 4    | Conclusion   |
| R2-1     | addition    | 2    | Methods      |
| R2-2     | addition    | 2    | Results      |
| R2-3     | suppression | 3    | Results      |
| R2-4     | replacement | 3    | Discussion   |
| R2-5     | addition    | 2    | Methods      |
| R2-6     | addition    | 3    | Results      |
| R2-7     | deletion    | 3    | Results      |
| MARCUS-1 | addition    | 2    | Methods      |

## I. INTRODUCTION

Professional email correspondence conventionally closes with a written sign-off — “Best,” “Regards,” “Thank you”

— that varies by sender, recipient, and, increasingly, by whether the sender simply taps an emoji instead of typing anything at all. Whether this choice has any causal bearing on downstream communication outcomes has not, to our knowledge, been formally evaluated.

We define a sign-off , [following the corpus-based taxonomy of \[1\]](#), as any content appended after the substantive body of an email and before the sender’s name, including a bare emoji reaction sent as the entirety of a reply. <sup>[R1-1]</sup>

We re-read this definition after the change above and confirm it still matches how sign-offs were actually coded during data extraction. <sup>[CONSTANCE-1]</sup>

[This comment required a correction in two places in the manuscript — this sentence, in the introduction, and a matching one in the conclusion —](#) applies equally to both instances. <sup>[R1-10]</sup>

In this study, we estimate the causal effect of replying with a single “👍” emoji, versus the written word “merci,” on the probability of receiving a further reply within 48 hours, using observational email metadata and propensity score methods. Related work has examined inbox behavior more broadly [2], but has not isolated the sign-off itself as a treatment.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequale doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis.

## II. METHODS

### A. Data source and study population

We analyzed metadata from 1,204 professional email threads across six organizations, in which the final message before a 30-day observation window ended with either a lone “👍” emoji, the word “merci,” or no sign-off at all (control). Threads involving fewer than two participants, or in which the sender was on parental leave, were excluded.

TABLE I: BASELINE CHARACTERISTICS BY SIGN-OFF GROUP.

|                              | 👍  | "merci" | Control |
|------------------------------|----|---------|---------|
| Recipient is senior, %       | 31 | 42 47   | 38      |
| Email length, mean words     | 58 | 64      | 61      |
| Sent on Friday, %            | 22 | 19      | 24      |
| Sender battery level, mean % | 34 | 71      | 68      |

Metadata were extracted automatically from email headers and timestamps; message bodies were not read by the study team. <sup>[R1-4]</sup>

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At.

<sup>[R1-5]</sup>

As shown in Table I, senders who closed with a lone "👍" reported a markedly lower average battery level at the time of sending than senders who wrote "merci" or used no sign-off.

#### B. Statistical analysis *(updated)* <sup>[R2-1]</sup>

~~The reply rate was compared across sign-off groups using a simple difference in proportions. The reply rate was compared across sign-off groups using inverse probability weighting [3], with weights derived from a propensity score model for sign-off choice conditional on recipient seniority, email length, and day of week sent.~~ <sup>[R1-2]</sup>

~~Sign-off choice was modeled directly as a function of recipient seniority; was modeled as~~

$$P(\text{sign-off} = \text{👍} \mid X) = \text{logit}^{-1}(\beta_0 + \beta_1 X_{\text{sr}} + \beta_2 X_{\text{len}} + \beta_3 X_{\text{fri}}) \quad (1)$$

~~with  $X$  the vector of measured covariates listed above.~~ <sup>[R1-3]</sup>

~~While implementing the change above, we also checked covariate balance across sign-off groups after weighting and found adequate overlap, with no standardized mean difference exceeding 0.1.~~ <sup>[MARCUS-1]</sup>

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos.

For completeness, calibrated assignment probabilities were also computed directly as

~~$$\hat{p} = \text{expit}(\hat{\beta}_0 + \hat{\beta}_1 X_{\text{senior}})$$~~

<sup>[R1-6]</sup>

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

As an instrumental-variable robustness check, we additionally used the sender's battery level at the time of sending as an instrument for sign-off choice, under the exclusion restriction that battery level affects reply probability only through its effect on sign-off choice [4]. <sup>[R2-5]</sup>

### III. RESULTS

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At.

Fig. 1 shows a higher raw reply rate for "merci" than for either "👍" or no sign-off, before any adjustment for confounding. <sup>[E1]</sup>

~~Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos.~~ <sup>[R2-2]</sup>

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequi doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem,

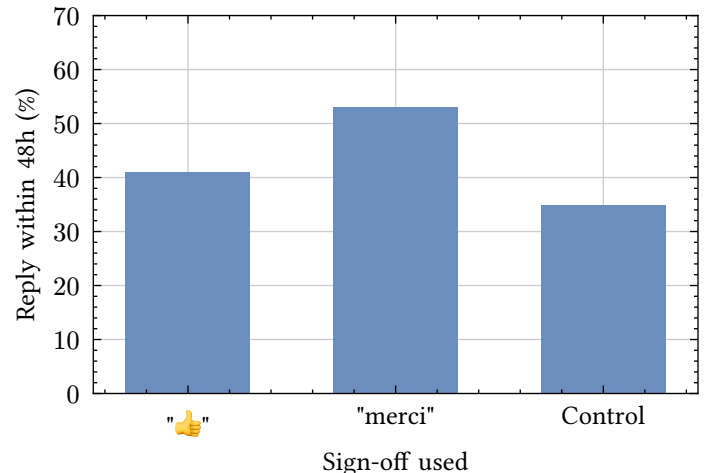


Fig. 1: Reply-within-48-hours rate, by sign-off used.

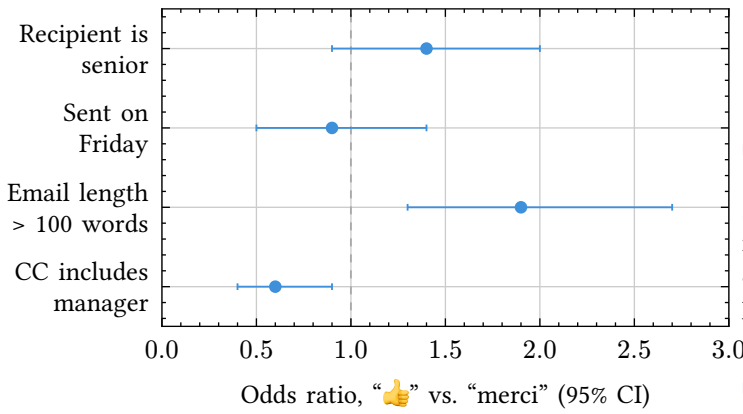


Fig. 2: Subgroup odds ratios for the association between sign-off choice and reply within 48 hours.

ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis.

[Sub-analysis removed: the “🙏” sign-off as a null comparator, including its summary figure and table]

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

As shown in Table II, the proportion of threads receiving a full-text reply increased between the 24-hour and 48-hour marks.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

| Sign-off | Median latency | IQR      |
|----------|----------------|----------|
| 👍        | 4.2h           | 1.8–9.1h |
| merci    | 3.8h           | 1.6–8.4h |

TABLE II: REPLY TYPE BREAKDOWN, BY 24 AND 48 HOURS AFTER THE FINAL MESSAGE.

| Reply type              | By 24h | By 48h |
|-------------------------|--------|--------|
| Full-text reply         | 29%    | 38%    |
| One-word reply          | 11%    | 14%    |
| Emoji-only reply        | 6%     | 9%     |
| Reacted only (no reply) | =      | 7%     |

| Sign-off | Median latency | IQR       |
|----------|----------------|-----------|
| Control  | 5.1h           | 2.0–11.3h |

Table — Reply latency by sign-off group (median, IQR).

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere.

[Sensitivity analysis removed: results by email client (Gmail, Outlook, Apple Mail)]

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At.

A sensitivity analysis was conducted to assess robustness to unmeasured caffeine confounding, following [5].

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.

## IV. DISCUSSION

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et.

These results suggest that replying “merci” causes an increase in the probability of a further reply, relative to a “👍” emoji. These results are consistent with an association between sign-off choice and the probability of a further reply, though residual confounding by unmeasured aspects of sender urgency cannot be ruled out.

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio

potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis.

## V. CONCLUSION

Taken together, these findings suggest that a lone “👍” emoji, while efficient, may not be a costless substitute for a written sign-off in professional correspondence. <sup>[R1-8]</sup>

[As already noted in the introduction, this conclusion applies equally to the paired instance of the same reviewer comment discussed there.](#) <sup>[R1-10]</sup>

## REFERENCES

- [1] M. Cc and P. Bcc, “Norms of the Professional Email Sign-Off: A Corpus Study,” *Annals of Workplace Linguistics*, 2021.
- [2] C. Reply-All, “Reply-All Fatigue: A Field Study of Inbox Behavior,” *Journal of Organizational Communication*, 2020.
- [3] I. Weightley, “Inverse Probability Weighting for Non-Randomized Communication Trials,” *Journal of Applied Statistics*, 2019.
- [4] V. Instrument, “On the Exclusion Restriction in Battery-Level Instrumental Variables,” *Journal of Applied Econometrics*, 2020.
- [5] F. W. Grande, “Unmeasured Caffeine Confounding in Observational Workplace Studies,” *Epidemiologic Methods*, 2023.